



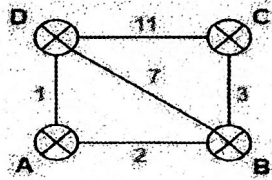
Internal Assessment – I

TERM: 3 rd October 2024 – 25 th January 2025	COURSE NAME: Advanced Computer Networks
DATE: 30-11-2024 TIME: 9.30 am-10.30 am	COURSE CODE: CIE552
MAX MARKS: 30	PORTIONS: L1 to L19



Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions. Marks: 15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Consider that an ISP is granted a block of addresses starting with 150.80.0.0/16 (65,536 addresses). The ISP needs to distribute these blocks to 2600 customers as follows: - The first group has 200 medium size businesses; each needs 128 addresses. - The second group has 400 small businesses; each needs 16 addresses. - The third group has 2000 households; each needs 4 addresses. - Design the sub-blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations..	L3	CO1	5
1.b	Calculate the checksum for the following ICMP packet: Type: Echo Request Identifier: 123 Sequence Number: 25 Message: Hello	L3	CO2	5
1.c	Describe the procedure in Distance Vector Routing Protocol for finding the least cost between the two nodes. Illustrate with an example given below 	L3	CO2	5
2.a	Describe IGMP format. Consider a situation in which there is only one host interested in a group, but the host is shutdown. The multicast router will never receive a leave report. Explain how IGMP handles this type of situation.	L2	CO2	5

2.b	An IP datagram has arrived with the following information in the header (in hexadecimal): 45 00 00 54 00 03 00 00 20 06 00 00 7C 4E 03 02 B4 0E 0F 02 - What is header length? - Is the packet fragmented? - How many more routers can the packet travel to? - What is the identification number of the packet? - What is the type of service?	L3	CO1	5
2.c	A computer receives a timestamp request from another computer at 2:34:20 P.M. The value of the original timestamp is 52,453,000. If the sender clock is 5 ms slow, what is the one-way time? Explain the various query message in ICMP and which query addresses the previous issue?	L3	CO2	5
3.a	Deliberate a scenario at GlobalTech Solutions, a company with several remote branch offices, the IT team has noticed that the corporate network is running out of available public IP addresses. The company's headquarters (HQ) has a large internal network with multiple subnets, each supporting dozens of devices. But the problem arises due to the limited IPv4 address availability and the increasing number of devices. Which technology can address their issue and what are the different ways of implementation?	L3	CO1	5
3.b	Explain the vulnerabilities of IPv4 in terms of security. How does IPv6 address these issues?	L2	CO1	5
3.c	Explain how to simulate the loose source route and strict source route option in a IP datagram. Illustrate and explain the procedure to find the route.	L3	CO2	5